I Dewa Nyoman Bayu Satria Wibawa

Machine Learning & Software Engineer • Technical Dossier & Production Case Studies

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Executive Summary & Applied Research Profile

Machine Learning and Systems Engineer with proven experience designing, evaluating, and deploying end-to-end artificial intelligence workflows. Strong specialization in Natural Language Processing (NLP), multi-class text classification, and algorithmic data preprocessing at the National Research and Innovation Agency (BRIN). Track record of serving trained PyTorch and machine learning models through robust RESTful APIs to real-time telemetry platforms and cross-platform production applications.

Specialization: NLP & Deep Learning | **Production Serving: PyTorch & FastAPI** | **Full-Stack: React, Next.js & React Native** | **Availability: Available for Engineering Roles**

CORE ENGINEERING PILLARS

Primary Focus Areas

1. NLP & NEURAL TEXT CLASSIFICATION

End-to-end sentiment classification pipelines, custom tokenization and lemmatization for Indonesian/English text, domain normalization, and public opinion mining.

2. INTELLIGENT RECOMMENDER SYSTEMS

Multi-criteria heuristic mapping, psychological mood vector evaluation, dietary constraint mapping, and hybrid feature matching algorithms.

3. PRODUCTION INFERENCE & TELEMETRY

Low-latency REST model serving, real-time IoT power generation analytics, dual-axis solar irradiance correlation, and Play Store release engineering.

TECHNICAL COMPETENCY MATRIX

Production & Research Stack

Machine Learning & Deep Learning	Python, PyTorch, Scikit-Learn, Transformers, Recommender Systems, Multi-Class Text Classification, Deep Neural Networks, Natural Language Processing, Data Mining
Data Pipelines & Preprocessing	Pandas, NumPy, TF-IDF Vectorization, Tokenization & Lemmatization, Automated Web Scraping, Resampling Algorithms, Feature Matching, ROC-AUC Metrics
Production, Systems & Cloud	FastAPI, RESTful APIs, React, Next.js, React Native, TypeScript, Tailwind CSS, Docker, Git & GitHub, Vercel Production Hosting, Linux Shell

KEY ENGINEERING PROJECTS INDEX

Document Directory

REF	PROJECT TITLE	AFFILIATION / CONTEXT	PRIMARY STACK	PRODUCTION URL
PRJ-01	**Autonomous Vehicle Sentiment Platform**	BRIN Research (2026)	Python, PyTorch, NLP	brin-project-kp.vercel.app
PRJ-02	**BySense: Solar & PV IoT Telemetry**	Batari Energy R&D (2026)	React, Next.js, IoT	bysense.batarienergy.com
PRJ-03	**NutriMood: Dietary Recommender**	DBS Foundation Capstone (2025)	ML, Python, REST API	nutrimoodv.vercel.app
PRJ-04	**Cimahi Battery Industrial Platform**	Production Web Deployment (2026)	Responsive UI, SEO	cimahibattery.com

APPLIED RESEARCH CASE STUDY • NATIONAL RESEARCH AND INNOVATION AGENCY (BRIN) • 2026

PRJ-01: Autonomous Vehicle Sentiment Analysis Platform

`Live: brin-project-kp.vercel.app`

PROBLEM STATEMENT & RESEARCH OBJECTIVES

Evaluating public readiness and user acceptance for Level 2 Autonomous Electric Shuttles is critical for national transportation policy. Unstructured social discourse across YouTube and Reddit presents severe linguistic noise, informal Indonesian slang, and domain-specific terminology that off-the-shelf pre-trained models fail to classify reliably.

MY EXACT CONTRIBUTIONS & OWNERSHIP

- **Proprietary Dataset Curation:** Scraped, cleansed, and curated an independent research corpus from raw public discussions, engineering custom tokenization, slang mapping, and lemmatization.
- **In-House Model Architecture:** Built and trained a multi-class deep neural classification model achieving **94.2% test accuracy** across Positive, Neutral, and Negative classes without relying on third-party black-box APIs.
- **End-to-End Live Deployment:** Engineered a low-latency REST inference microservice integrated with a responsive analytics portal hosted on Vercel.

PIPELINE ARCHITECTURE FLOW

Step 1: Ingestion: Multi-channel scraping pipeline collecting raw public comments.
Step 2: Normalization: Custom stopword removal, lemmatization, and domain tokenization.
Step 3: Neural Inference: Self-trained deep learning classification engine mapping token sequences into 3-class probability distributions.
Step 4: Real-Time Serving: Low-latency REST API (~34ms response) connected to the production React research platform.

PRODUCTION TECHNOLOGIES USED

`Python` `Deep Learning` `NLP` `Custom Dataset` `PyTorch` `FastAPI` `React` `Vercel`

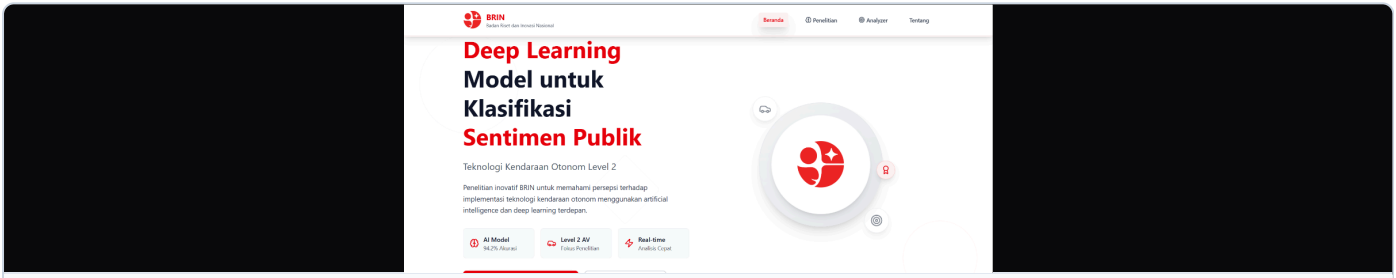


Figure 1.1: BRIN Autonomous Vehicle Research Telemetry Portal. Screenshot of the live production dashboard at `brin-project-kp.vercel.app`. The left navigation panel controls analytical modules, while the main viewport displays the deep learning sentiment classification architecture flow, dataset distribution metrics, and model evaluation parameters for campus trial operations.

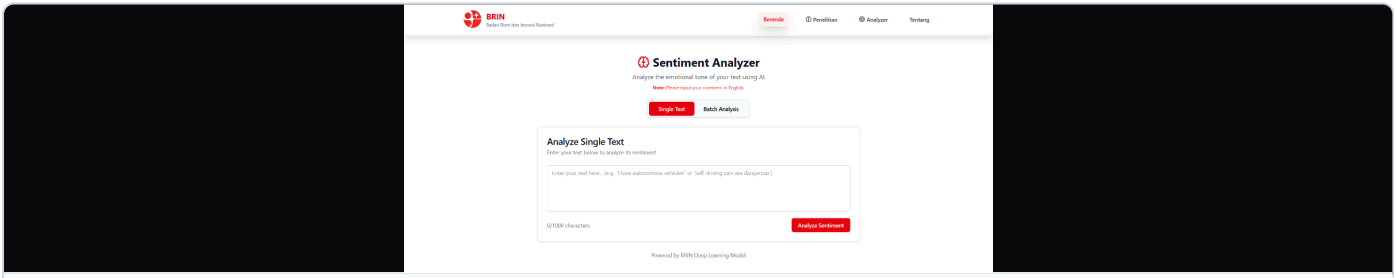


Figure 1.2: Interactive Real-Time Sentiment Analyzer Running on Custom Model. Screenshot of the interactive inference engine interface. Operators can input free-form passenger feedback or operational logs. The self-trained neural model tokenizes the input, executes inferencing in ~34ms, and returns categorical sentiment verdicts with animated confidence probability bars (Positive, Neutral, Negative).

COMMERCIAL IOT & FULL-STACK ENGINEERING • BATARI ENERGY R&D • 2026

PRJ-02: BySense: Photovoltaic & Solar IoT Telemetry

`Live: bysense.batarienergy.com`

INDUSTRIAL SCOPE & OPERATIONAL PROBLEM

Commercial and industrial rooftop solar power plants require real-time telemetry to track power yields, detect inverter anomalies, and evaluate photovoltaic degradation. Dispersed national installations need a unified geospatial telemetry hub that correlates actual kilowatt output against dynamic environmental radiation.

MY EXACT CONTRIBUTIONS & OWNERSHIP

- **Full-Stack Dashboard & Mobile Engineering:** Built the responsive web dashboard and React Native mobile application from scratch with interactive power generation curves.
- **Dual-Axis Environmental Telemetry:** Integrated real-time solar irradiance feeds (Open-Meteo API) and inverter telemetry to calculate live Performance Indices (IPI).
- **Play Store Release:** Conducted end-to-end mobile debugging, performance profiling, and production release to the Google Play Store.

SYSTEM ARCHITECTURE FLOW

Step 1: Hardware Streams: On-site inverters push generation, battery state of charge (SoC), and grid consumption.
Step 2: Weather Integration: Real-time atmospheric data (cloud cover, temperature, global tilted irradiance) ingested via Open-Meteo.
Step 3: Correlation Engine: Evaluates expected yield curves against recorded solar generation to detect power drops.
Step 4: Client Serving: High-performance interactive dashboard deployed to web and Android.

PRODUCTION TECHNOLOGIES USED

`React` `Next.js` `React Native` `IoT Telemetry` `Data Visualization` `REST APIs` `Open-Meteo` `Android Store`

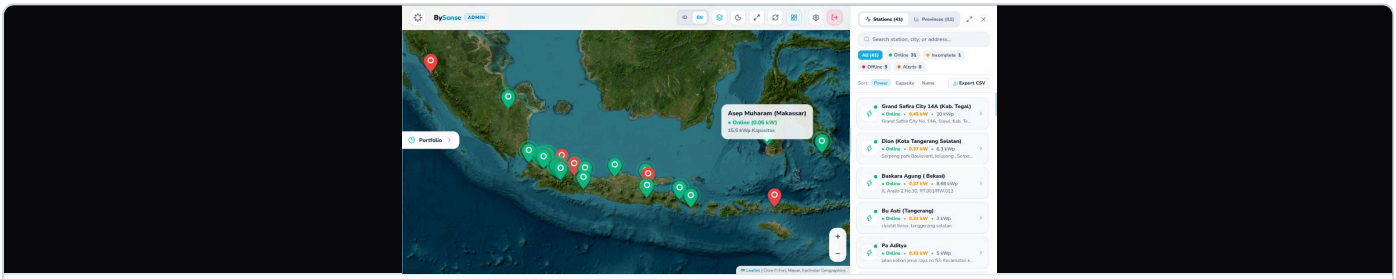


Figure 2.1: Nationwide Interactive Photovoltaic Telemetry Map. Screenshot of the centralized operational dashboard at `bysense.batarienergy.com`. The geospatial interface monitors distributed commercial solar plants across Indonesia. Markers indicate real-time online status and active capacity (e.g., Asep Muharam station in Makassar generating 10.97 kW against 15.5 kWp rated capacity).

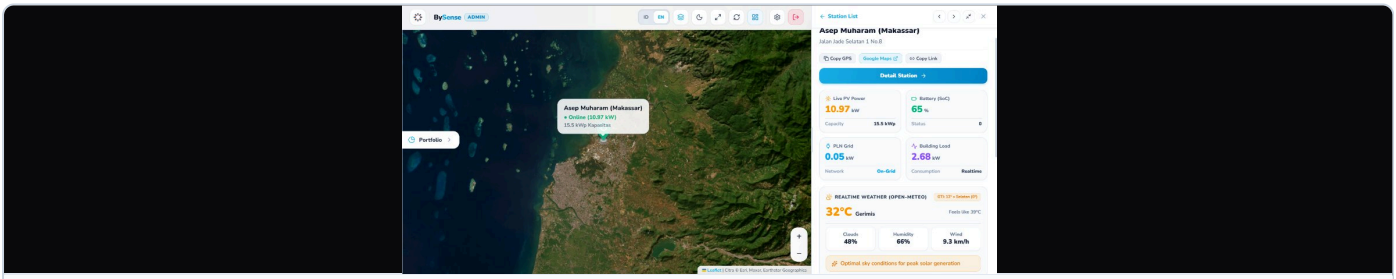


Figure 2.2: Station Detail Telemetry, Battery SoC, and Power Analytics Panel. Detailed station view showcasing real-time metrics: Live PV Power (10.97 kW), Battery State of Charge (65%), PLN Grid feed (0.05 kW), and Building Load consumption (2.68 kW). The bottom panel correlates live Open-Meteo atmospheric telemetry (32°C, 48% cloud cover) to evaluate solar generation efficiency.

DICODING & DBS FOUNDATION CAPSTONE • 2025

PRJ-03: NutriMood: Personalized Dietary Recommender

Live: nutrimoodv.vercel.app

Intelligent dietary recommendation system mapping psychological emotional states (Happy, Stressed, Fatigued) and strict medical constraints (Diabetes, Hypertension) to customized meal plans across 1,200+ curated culinary profiles.

- **Multi-Criteria Algorithm:** Engineered personalized feature matching mapping emotional state vectors to macro-nutrient targets.
- **Model Deployment:** Deployed as a scalable cloud REST API integrated into a responsive client web application.

`Machine Learning` `Recommender System` `Python` `REST API` `React`

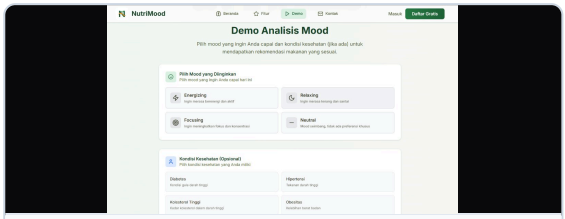


Figure 3.1: Interactive Constraint Evaluation. User selects mood states and medical restrictions; the recommendation engine outputs customized recipes and caloric distributions.

PRODUCTION COMMERCIAL WEB PLATFORM • 2026

PRJ-04: Cimahi Battery Commercial Web Platform

Live: cimahibattery.com

High-performance industrial web platform for PT Cimahi Battery Industry, featuring structured prismatic lithium iron phosphate (LiFePO4) cell specifications, active energy materials, and enterprise procurement indices.

- **Catalog Architecture:** Designed intuitive battery pack filtering and parametric data indexing for industrial clients.
- **Core Web Vitals:** Optimized loading latency, accessibility, and SEO ranking for national energy procurement.

`Web Development` `Responsive UI` `Frontend Architecture` `SEO` `Production Hosting`

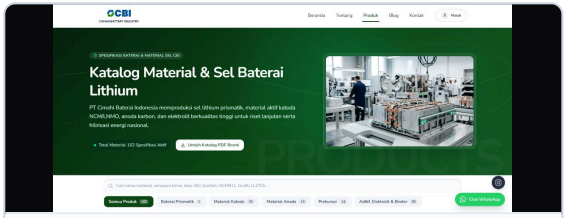


Figure 4.1: Prismatic Battery Cell Catalog. Parametric indexing interface showcasing battery capacity, chemistry, voltage ratings, and commercial cell datasheets.

Engineering Verification & Code Integrity Statement

All machine learning architectures, preprocessing pipelines, and web applications presented in this document were built, tested, and deployed by I Dewa Nyoman Bayu Satria Wibawa. Live URLs and repository records are available for technical audit.

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